NEXT CHAPTER @ SMITH PUBLIC LIBRARY

Ratios Are Everywhere

Session 3 — Math Part B: Ratios & Percentage Applications

Monday Evenings • 5:30–7:30 PM • Library Conference Room

Tonight's Agenda

1	Quick recap of last week	*10 min*
2	Ratios in real life: concept	*15 min*
3	Ratio guided practice	*25 min*
4	Percentage applications: discounts, tips, tax	*20 min*
5	Two worked examples together	*25 min*
6	Mixed independent practice	*20 min*
7	Wrap-up & take-home preview	*5 min*

Quick Recap: Last Week

Order of operations:

Parentheses → Exponents → Multiply/Divide → Add/Subtract

Same number, three costumes:

1/2 = 0.5 = 50% — fractions, decimals, and percentages are the same value in different forms.

Ratios Are Everywhere

Recipes

“2 cups flour to 1 cup sugar” is a 2:1 ratio — double the recipe, keep the ratio.

Pay Rates

“$18 for 2 hours” is a rate — the same idea as a ratio, applied to money and time.

Mixing & Scaling

Paint colors, cleaning solutions, and map scales all use ratios to stay proportional.

How to Solve a Ratio Problem

The Scale Factor Method

Step 1: Find what number the known side was multiplied by.

Step 2: Multiply the other side by that same number.

Example: A recipe uses 2 cups flour to 1 cup sugar. How much sugar for 6 cups flour?

2 was multiplied by 3 to get 6 → so multiply 1 by 3 too → 3 cups sugar.

Your Turn: Solve These Ratios

GUIDED PRACTICE

1. A recipe uses 3 cups flour to 2 cups sugar. How much sugar for 9 cups flour?

2. You earn $45 for 3 hours of work. How much do you earn for 5 hours?

3. A map shows 1 inch = 10 miles. How many miles is 4.5 inches?

4. A paint mix is 2 parts blue to 5 parts white. How much white for 6 parts blue?

Let's Check Your Answers

1. 3:2 scaled by 3 → 9 cups flour needs 6 cups sugar

2. $45 ÷ 3 = $15/hour → 5 hours = $75

3. 1 in = 10 mi → 4.5 × 10 = 45 miles

4. 2:5 scaled by 3 → 6 parts blue needs 15 parts white

Percentages in Real Life

Discounts

“25% off” means you pay 75% of the original price.

Tips

A 20% tip on \$30 is $0.20 \times 30 =$ \$6.

Sales Tax

8% tax on \$50 adds $0.08 \times 50 =$ \$4, for a total of \$54.

Interest

Simple interest works the same way — a percentage of a starting amount.

Worked Example: Tip Calculation

Your restaurant bill is $42. You want to leave an 18% tip. What's the total?

Step 1: Find 18% of $42 → 0.18 × 42 = $7.56

Step 2: Add the tip to the bill → $42 + $7.56 = $49.56

Total: $49.56

Worked Example: Discount + Tax

A $80 jacket is 30% off, then 8% sales tax is added. What's the final price?

Step 1: Find the discount → $0.30 \times 80 = \$24$

Step 2: Sale price → $\$80 - \$24 = \$56$

Step 3: Find the tax → $0.08 \times 56 = \$4.48$

Step 4: Final total → $\$56 + \$4.48 = \$60.48$

Final price: $60.48

Independent Practice: Mixed Review

ON YOUR OWN

1. $18 - 4 \times 3 = ?$

2. Write 7/10 as a percentage.

3. A recipe uses 4 cups water to 3 cups rice. How much water for 12 cups rice?

4. A $25 shirt is 20% off. What's the sale price?

5. You tip 15% on a $60 bill. How much is the tip?

Mixed Review Answers

1. $18 - 4 \times 3 = 18 - 12 = 6$

2. $7/10 = 0.7 \rightarrow 70\%$

3. 4:3 scaled by 4 → 12 cups rice needs 16 cups water

4. 20% of \$25 = \$5 → sale price \$20

5. 15% of \$60 = \$9 tip

Math Block: Complete.

See you next Monday — we build on this with algebra.

Before next week:
Try tonight's take-home practice. Want more? GED.com's free test preview tutorials (tools.ged.com) are always open, in English or Spanish.